

MRC Lecture 6 — Derivation Guide 2

From One Equation to an Architecture

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Sources: Methods for Molecular Recognition Computing, Supplementary Note S2; Computational Convergence, Supplementary Note S1 (CDR multi-head structure, emergent antibody–antigen bilinearity).

1. Motivation

WB1 proved that the convergence equation $P(j) = \exp(s_j/\tau) / \sum_l \exp(s_l/\tau)$ is the unique selection function for competitive binding at equilibrium. But the equation takes abstract scores s_j as input. This guide shows where the scores come from in a molecular system, and that once you supply them, the **architecture assembles itself**. The bilinear decomposition of binding energy turns the convergence equation into the transformer attention equation; the dual-encoder, cross-attention structure follows; multi-head attention emerges as the image of an antibody's multiple contact regions; and unequal target concentrations enter as an additive bias term — exactly, from the physics, not as an approximation.

The lesson: the SFM architecture is not designed. It is what the convergence equation looks like when the scores are molecular compatibility scores.

2. The Score Is a Dot Product

From Lecture 5, the binding energy decomposes bilinearly. In the favorable (more negative ΔG) convention, the compatibility score is a dot product between an agent feature vector $\mathbf{q}$ and a target feature vector $\mathbf{k}_j$:

$$s_j = \mathbf{q} \cdot \mathbf{k}_j$$

There are two regimes:

- **Given bilinearity** (nucleic acids): $\mathbf{q}$ and $\mathbf{k}_j$ are physical, interpretable features in raw sequence space — the position weight matrix and the one-hot DNA encoding (transcription factors), or the nearest-neighbor parameters and the sequence histogram (SantaLucia). No encoder training is needed for the form.
- **Emergent bilinearity** (proteins): $\mathbf{q} = \text{Encoder}_A(\text{agent})$ and $\mathbf{k}_j = \text{Encoder}_T(\text{target}_j)$ are learned embeddings (ESM-2, AntiBERTy). The dot product exists only after the encoder discovers the right representation.

In both regimes, the score is a dot product. That single fact is what generates the architecture.

3. Substituting Into the Convergence Equation

Replace the abstract score s_j in the convergence equation with the bilinear form $\mathbf{q} \cdot \mathbf{k}_j$:

$$P(j) = \frac{\exp(\mathbf{q} \cdot \mathbf{k}_j/\tau)}{\sum_l \exp(\mathbf{q} \cdot \mathbf{k}_l/\tau)}$$

Write the transformer attention weight (Vaswani et al., 2017) directly beneath it:

$$A_j = \frac{\exp(\mathbf{q} \cdot \mathbf{k}_j/\tau)}{\sum_l \exp(\mathbf{q} \cdot \mathbf{k}_l/\tau)}$$

These are the same equation. The convergence equation with a bilinear molecular score *is* the transformer attention mechanism. The attention mechanism was not borrowed from AI and applied to biology — it is what the physics of competitive binding produces when the binding energy is bilinear.

4. Worked Example: The PWM Score Is a Dot Product

To make the dot product concrete, take a transcription factor with a 6-position binding motif. Its position weight matrix $\mathbf{W}$ (rows A, C, G, T; columns positions 1–6) is:

	p1	p2	p3	p4	p5	p6
A	3	-1	0	1	-1	3
C	-1	4	-1	0	3	-1
G	0	-1	2	3	-1	0
T	-1	0	-1	-1	1	-1

Scoring the consensus site ACGGCA

The PWM score is the sum of the per-position contributions for the bases actually present:

$$S = W_{1,A} + W_{2,C} + W_{3,G} + W_{4,G} + W_{5,C} + W_{6,A} \\ = 3 + 4 + 2 + 3 + 3 + 3 = \boxed{18}$$

This sum is a dot product

Encode the DNA site as a one-hot vector $\mathbf{x}$ (flattening the 6×4 one-hot matrix into 24 entries) and flatten the PWM into the corresponding 24-entry vector $\mathbf{w}$. Then:

$$S = \mathbf{w} \cdot \mathbf{x} = \sum_{p,b} W_{p,b} X_{p,b} = 18$$

The one-hot $\mathbf{x}$ selects exactly one weight per position; the dot product sums them. **The PWM lookup-and-sum IS the dot product $\mathbf{q} \cdot \mathbf{k}$** , with the flattened PWM playing the role of the query $\mathbf{q}$ and the one-hot DNA playing the role of the key $\mathbf{k}$.

Contrast: a non-consensus site

Score TGCATG:

$$S = W_{1,T} + W_{2,G} + W_{3,C} + W_{4,A} + W_{5,T} + W_{6,G} = -1 + (-1) + (-1) + 1 + 1 + 0 = -1$$

The consensus site scores 18, the non-consensus site scores -1 . Feeding these into the convergence equation, the consensus site dominates the binding probability — exactly as a TF concentrates on its motif, and exactly as an attention head concentrates on its most compatible key.

5. The Architecture Assembles Itself

The bilinear score $\mathbf{q} \cdot \mathbf{k}_j$ dictates the architectural components:

1. **Something must produce $\mathbf{q}$** — an encoder for the agent molecule.
2. **Something must produce $\mathbf{k}_j$** — an encoder for the target molecule.
3. Since agents and targets are different molecule types (antibody vs. antigen, TF vs. DNA), these are **two separate encoders** — *dual encoders*.
4. **The interaction is a dot product** between the two encoder outputs — *cross-attention*.
5. **The selection is the softmax over the pool** — the convergence equation.

None of these components is a design choice. Each is forced by the bilinear structure of the score and the uniqueness of the convergence equation. The skeleton — dual encoders, dot-product cross-attention, softmax — is prescribed.

6. Given vs. Emergent Bilinearity

The same architecture serves both regimes; only what fills the encoder slots changes.

	Given bilinearity	Emergent bilinearity
Domains	TF–DNA, miRNA–mRNA, CRISPR	Ab–Ag, TCR–pMHC, enzyme–substrate, drug–target

	Given bilinearity	Emergent bilinearity
q, k	Physical / interpretable (PWM, NN params, one-hot)	Learned embeddings (ESM-2, AntiBERTy, DNABERT-2)
Encoder	Identity or lookup; form known from physics	Pretrained language model; form learned from data
Score basis	Raw sequence space	Learned embedding space
Architecture	dual encoders + dot product + softmax	dual encoders + dot product + softmax

The architecture is identical. The convergence equation does not care whether the dot product lives in raw sequence space or a learned embedding space — it only requires that the score be a dot product. This is why the same SFM code trains across both regimes (Lecture 8).

7. Multi-Head Attention = Multiple Contact Regions

A single dot product $\mathbf{q} \cdot \mathbf{k}_j$ produces one scalar compatibility score. But an antibody does not contact its antigen at a single point — its six **CDR loops** (CDRH1, H2, H3, CDRL1, L2, L3) each contact a different region of the epitope surface. The total binding energy is a sum over these distinct contact regions.

This is exactly what **multi-head attention** computes. With H heads, the agent and target are each projected into H separate subspaces, a dot product is computed in each, and the results are combined:

$$\text{MultiHead}(\mathbf{q}, \mathbf{k}) = \text{Concat}(\text{head}_1, \dots, \text{head}_H), \quad \text{head}_h = \mathbf{q}^{(h)} \cdot \mathbf{k}^{(h)}$$

Each head attends to a different contact region — the architectural image of multiple CDR loops contacting multiple epitope patches. For **emergent** antibody–antigen binding, the multi-head structure is not optional: distributed contacts across a 3D interface cannot be captured by a single scalar dot product. The CDR loops *are* the attention heads. (Computational Convergence, Supplementary Note S1, develops this correspondence for adaptive immune recognition.)

8. Concentration Enters as a Bias Term

The convergence equation as written assumes all targets are present at equal concentration. When concentrations differ — for instance, an enzyme choosing among substrates at unequal concentrations — the physics adds a term, **exactly**, with no approximation.

At thermal equilibrium, the probability of binding target j is weighted by both the Boltzmann factor and the target's concentration $[S_j]$:

$$P(\text{bind } j) = \frac{[S_j] \exp(-\Delta G_j/RT)}{\sum_l [S_l] \exp(-\Delta G_l/RT)}$$

Using $[S_j] = \exp(\ln[S_j])$, fold the concentration into the exponent:

$$P(\text{bind } j) = \frac{\exp(-\Delta G_j/RT + \ln[S_j])}{\sum_l \exp(-\Delta G_l/RT + \ln[S_l])}$$

Define the **effective score**:

$$s_j^{\text{eff}} = \frac{\mathbf{q} \cdot \mathbf{k}_j}{\tau} + \ln[S_j]$$

The concentration enters as an **additive bias** in the attention logit. This is exact — a direct consequence of the Boltzmann distribution, not a modeling heuristic. Architecturally, it is a per-target additive bias added to the dot-product logits before the softmax, identical to how positional or masking biases are added in standard transformers. (Methods for Molecular Recognition Computing, Supplementary Note S2, derives this for the enzyme–substrate SFM.)

Consequence. Systems with strongly varying target concentrations (enzyme–substrate, drug–target at varying doses) require the $\ln[S]$ bias term; systems where the pool is effectively equimolar (retrieval over a curated candidate set) do not. The architecture accommodates both: the bias term is present or set to zero. Either way it is prescribed by the physics, not invented.

9. Summary

Concept	Result	Significance
Score is a dot product	$s_j = \mathbf{q} \cdot \mathbf{k}_j$	Given (raw) or emergent (learned)
Substitution	$P(j) = \exp(\mathbf{q} \cdot \mathbf{k}_j / \tau) / Z$	Convergence equation = attention
PWM example	consensus ACGGCA $\rightarrow$ score = 18	PWM lookup-and-sum IS a dot product
Architecture	dual encoders + cross-attention + softmax	Prescribed, not designed
Given vs. emergent	same architecture, different encoder fill	One code, all domains
Multi-head	H heads = H contact regions	CDR loops are attention heads
Concentration	$s_j^{\text{eff}} = \mathbf{q} \cdot \mathbf{k}_j / \tau + \ln[S_j]$	Exact additive bias from Boltzmann

10. Checkpoint Questions

Q1. Using the PWM in §4, score the site ACGGCC (consensus with the last base changed from A to C). By how much does the score drop, and what does this say about position 6’s specificity?

Answer: The only change is position 6: A scores $W_{6,A} = 3$, C scores $W_{6,C} = -1$. New score $S = 18 - 3 + (-1) = 14$. A drop of 4. Position 6 strongly prefers A (the gap between A at +3 and C at -1 is the maximum 4-point swing in that column), so it is a high-specificity “anchor” position. Feeding 18 vs. 14 into the convergence equation with $\tau = 1$, the consensus is $\exp(4) \approx 55$ times more probable than this single-mutant — a steep penalty for disrupting an anchor position.

Q2. An enzyme has two substrates with equal intrinsic compatibility ($\mathbf{q} \cdot \mathbf{k}_1 = \mathbf{q} \cdot \mathbf{k}_2$) but substrate 1 is present at $100\times$ the concentration of substrate 2. With $\tau = 1$, what is the ratio of binding probabilities?

Answer: The effective scores differ only by the concentration bias: $s_1^{\text{eff}} - s_2^{\text{eff}} = \ln[S_1] - \ln[S_2] = \ln(100) = 4.605$. The probability ratio is $P_1/P_2 = \exp(4.605) = 100$. The enzyme binds substrate 1 $100\times$ more often — entirely due to concentration, exactly tracking the abundance ratio. This is the expected mass-action result, recovered exactly from the Boltzmann form because the concentration term enters as $\ln[S]$ in the logit.

Q3. Why must antibody–antigen binding use multi-head attention, while a simple single-position recognition (one nucleotide contact) could in principle use a single head?

Answer: A single attention head computes one scalar dot product — one compatibility channel between a single agent representation and a single target representation. Antibody–antigen binding involves a 3D interface with ~ 15 – 25 residue contacts distributed across six CDR loops, where different loops contact different epitope regions and the total energy is a sum over these spatially distinct channels. A single dot product cannot represent multiple independent contact channels; multi-head attention provides one channel per head, matching the CDR-loop structure. A single-nucleotide contact has only one channel and could be captured by one head (though in practice even nucleic acid systems use multiple heads to capture context). The number of heads needed scales with the number of independent contact regions.

Q4. A student says: “If the architecture is fully prescribed, then building an SFM is trivial — just plug in the equation.” What is wrong with this, and what is actually left to do?

Answer: What is prescribed is the *skeleton*: dual encoders, dot-product cross-attention, softmax, optional concentration bias. What is *not* prescribed — and what determines whether the SFM works — is everything that fills the skeleton: which pretrained encoders to use for each molecule type, how to curate and clean the training pairs, how to design hard negatives (the decisive choice, per WB3 and Lecture 9), how to split data to prevent leakage, and how to evaluate. These are domain-expert decisions, not architectural ones. The prescription removes the architecture-design debate; it does not remove the scientific work of building a good model. That is precisely why the course spends Lectures 8–9 on the build process and the final project on doing it.

11. Connection to Future Lectures

- **Lecture 6 WB3:** With the architecture in hand, the training objective is derived. InfoNCE is the maximum-likelihood estimator of the convergence equation; the symmetric form follows from binding reciprocity.
- **Lecture 7:** The five architecture conditions formalize the assembly shown here. The “prescribed vs. domain-specific” table separates the forced skeleton (this guide) from the free choices (encoders, data, hard negatives). Level 3 alignment is defined.
- **Lecture 8:** The given/emergent table (§6) is developed across all ten domains, with the six prototyped SFMs showing the identical architecture trained on different molecular systems.
- **Lecture 9 (build):** Students implement exactly this architecture using Claude Code with the CALM reference. The concentration bias (§8) is relevant for enzyme/drug–target projects; the multi-head structure (§7) is relevant for all protein-interface projects.